

DDoS Attack Detection in Consumer IoT-Based Healthcare Systems Using Improved Off-Policy Proximal Policy Optimization and Generative Adversarial Network

Jing Yang¹, Graduate Student Member, IEEE, Vijay Govindarajan², Lip Yee Por³, Senior Member, IEEE, Zaffar Ahmed Shaikh⁴, Qin Xin⁵, Pronaya Bhattacharya⁶, Member, IEEE, Abdullah Ayub Khan, and Yiyu Wang⁷

Abstract—Distributed denial of service (DDoS) attacks threaten the reliability of consumer IoT-based healthcare systems. However, traditional DDoS detection methods often struggle with feature selection, imbalanced datasets, generalizability, and hyperparameter optimization. This paper proposes a novel method for detecting DDoS attacks in healthcare environments. To enhance feature selection and handle imbalanced data, we employ the off-policy proximal policy optimization (PPO) algorithm, which improves sample efficiency by leveraging past experiences in reinforcement learning. SHAP (SHapley Additive exPlanations) values are integrated for precise feature identification, while augmented rewards for underrepresented classes help mitigate data imbalance. An improved generative adversarial network (GAN) is used for online data augmentation, selectively excluding gradients from critical batch elements to generate more diverse and applicable results. Furthermore, the Bayesian Optimization Hyperband (BOHB) method accelerates hyperparameter tuning by combining Bayesian optimization

with Hyperband's scaling capabilities. Our model is evaluated on four datasets: KDDcup99, ISCX UNB, CICDDOS, and DARPA, achieving superior performance over state of the art methods, with accuracy and F-measure values of (90.498%, 89.286%), (93.186%, 91.896%), (92.403%, 90.285%), and (92.849%, 92.790%), respectively. It demonstrates the model's efficiency in delivering timely predictions, supporting its practical deployment in real-time healthcare cybersecurity systems.

Index Terms—DDoS attack, proximal policy optimization, GAN, consumer healthcare.

I. INTRODUCTION

DDoS attacks pose significant risks to networks, potentially disabling them entirely [1], [2]. This often leaves system administrators preoccupied with mitigating these attacks, possibly overlooking other malicious activities co-occurring. Recent trends indicate a substantial escalation in both the frequency and magnitude of these attacks [3]. Moreover, the IoT proliferation has introduced a new wave of IoT-driven botnets that threaten Internet stability through DDoS onslaughts. This includes the Internet of Medical Things (IoMT), which is increasingly essential in delivering health services. According to Somasundaram and Thirugnanam [4], DDoS stands out as particularly hazardous within these networks. Also, numerous organizations integrate additional IoT devices within their systems without effectively filling the loopholes of the existing security issues, thus expanding the scope and scale of IoT botnet attacks. So, it becomes crucial to design methods of countering such threats, especially in systems that handle sensitive information, such as healthcare systems [5].

Recent advances have introduced various machine learning (ML) [6] and deep learning (DL) methods [7] for DDoS attack detection. While DL offers advantages over traditional ML, challenges remain in feature extraction, class imbalance, generalizability, and hyperparameter tuning. These aspects are often underexplored in existing studies. Feature selection is essential for improving DL model efficiency by removing irrelevant variables. Although DL can perform automatic feature extraction, targeted selection helps prevent overfitting, reduce

Received 30 March 2025; revised 21 June 2025 and 3 August 2025; accepted 25 August 2025. Date of publication 1 September 2025; date of current version 8 December 2025. This work was supported in part by the National Science and Technology Council, Taiwan, under Grant NSTC-114-2224-E-027-001; private funding (PV009-2023); and in part by the KW IPPP (Research Maintenance Fee) Individual/Centre/Group Universiti Malaya, Malaysia, under Grant RMF1506-2021. (Corresponding authors: Yiyu Wang; Lip Yee Por.)

Jing Yang and Lip Yee Por are with the Center of Research for Cyber Security and Network, Faculty of Computer Science and Information Technology, Universiti Malaya, Kuala Lumpur 50603, Malaysia (e-mail: s2147529@siswa.um.edu.my; porlip@um.edu.my).

Vijay Govindarajan is with the Distribution and Supply Technology, Expedia Group, Seattle, WA 98119 USA (e-mail: vigovindaraja@expediagroup.com).

Zaffar Ahmed Shaikh is with Department of Computer Science and Information Technology, Benazir Bhutto Shaheed University Lyari, Karachi 75660, Pakistan, and also with the School of Engineering, École Polytechnique Fédérale de Lausanne, 1015 Lausanne, Switzerland (e-mail: zashaikh@bbsul.edu.pk).

Qin Xin is with the Faculty of Science and Technology, University of the Faroe Islands, 100 Tórshavn, Faroe Islands (e-mail: qinx@setur.fo).

Pronaya Bhattacharya is with the Department of Computer Science and Engineering, Amity School of Engineering and Technology, and the Research and Innovation Cell, Amity University, Koltata 700135, India (e-mail: pbhattacharya@kol.amity.edu).

Abdullah Ayub Khan is with the Department of Computer Science, Bahria University Karachi Campus, Karachi 75260, Pakistan (e-mail: abdullahayub.bukc@bahria.edu.pk).

Yiyu Wang is with the J.E. Cairnes School of Business and Economics, University of Galway, Galway, H91 TK33 Ireland (e-mail: y.wang32@universityofgalway.ie).

Digital Object Identifier 10.1109/TCE.2025.3605098

training time, and enhance adaptability. However, current selection methods often perform poorly in dynamic environments. Class imbalance is typically handled through data-level strategies such as SMOTE [8] and NearMiss, which may cause overfitting or information loss, or through algorithm-level approaches including cost-sensitive learning, ensemble models, and threshold tuning. These methods emphasize minority classes but may reduce performance on majority class predictions [9].

Advanced deep reinforcement learning (DRL) addresses challenges in feature selection and class imbalance by applying reward mechanisms that prioritize critical yet often overlooked attributes. The integration of explainable artificial intelligence (AI) techniques, such as SHAP values, enhances the transparency of this process. SHAP quantifies the contribution of each feature to the model's output, ensuring focus on attributes with high explanatory value. This improves feature selection and promotes trust in AI decisions. DRL further improves class balance by boosting predictions for under-represented categories while maintaining overall performance. However, both RL and DRL face notable limitations. RL often requires substantial interaction with the environment and may exhibit instability or slow convergence due to noisy or sparse rewards. DRL is powerful but has high computational requirements and sample inefficiency, particularly when it cannot effectively reuse off-policy data. Off-policy PPO improves DRL by using replay buffers to store previous interactions. This increases sampling efficiency and lowers the reliance on constant new data. These enhancements enable DRL to adapt in dynamic settings, outperforming static conventional models and maintaining robust performance across various applications, including DDoS attack detection.

Data augmentation improves classifier performance by artificially expanding the training set, enhancing class balance, and supporting model generalization [10]. It is typically categorized into offline and online methods. Offline augmentation enlarges the dataset but increases computational load and training time. In contrast, online augmentation introduces synthetic samples during training without expanding the dataset or requiring additional storage. Recent advances in online augmentation leverage GANs, which provide strong generalization. However, GANs often face challenges such as training instability and mode collapse. To address these limitations, this study proposes a modified GAN that excludes the gradients of key batch members, helping stabilize training and enhance sample diversity for healthcare applications [11].

Selecting appropriate hyperparameters is crucial for optimizing deep learning models and preventing both overfitting and underfitting [12]. Due to the large search space, this task is complex and requires effective optimization strategies to address. Various optimization strategies address this issue, including grid search, Bayesian optimization, evolutionary algorithms, and Hyperband. Grid search is thorough but inefficient for large sets. Bayesian optimization offers efficient sampling but may converge on local optima. Evolutionary algorithms can handle complex spaces, but they are resource-intensive. Hyperband improves efficiency by focusing on promising configurations early [13]. This paper utilizes

BOHB for hyperparameter optimization, which combines the predictive efficiency of Bayesian optimization with the resource-saving strategy of Hyperband. This method provides a scalable solution for tuning hyperparameters in complex tasks such as anomaly detection [14].

This paper proposes a DDoS attack detection model tailored for healthcare systems, incorporating an off-policy PPO algorithm to facilitate feature selection and mitigate class imbalance. An enhanced GAN is introduced for online data augmentation, and BOHB is employed for efficient hyperparameter tuning. The main contributions of this study, which are considered original within the field, are summarized as follows:

- We proposed an off-policy PPO algorithm that leverages past experiences to train policies in accordance with reinforcement learning principles. By integrating SHAP values, this approach enhances feature selection, allowing the model to focus on the most relevant features and improve overall detection performance.
- We introduced an updated GAN that selectively discards gradients of critical batch members during training, enabling the generation of diverse and relevant samples to augment the training set without expanding the database.
- We designed the BOHB method for hyperparameter tuning, combining the accuracy of Bayesian optimization with the speed of Hyperband to accelerate convergence by efficiently discarding suboptimal configurations.

II. RELATED WORK

A. Machine Learning

A range of ML approaches have been employed for DDoS detection, utilizing algorithms such as decision trees, support vector machines, random forests, and k-nearest neighbors. Feng et al. [15] proposed an enhanced KNN algorithm using a k-dimensional tree for efficient source classification based on IP address risk levels. Tang et al. [16] adopted a pairwise learning-to-rank (LtR) algorithm to prioritize flow entries in SDN, aiming to mitigate low-rate DDoS attacks. Xu et al. [17] designed a federated learning framework incorporating evolutionary game theory and auction mechanisms for bandwidth optimization under mobile edge network attacks. Zhao et al. [18] introduced a machine learning-based scheduling system that encodes victim-defined traffic preferences without full knowledge of attack patterns. Mirnajafizadeh et al. [19] proposed a sublinear ML-based flow prioritization framework for improving DDoS detection accuracy. Chandak and Ray [1] focused on resource-constrained environments, integrating edge nodes with a feature-selected SVM optimized using an information gain ratio. Altulaihan et al. [6] developed an IoT intrusion detection system using supervised classifiers (DT, RF, KNN, and SVM) along with correlation-based feature selection and genetic algorithms. Setia et al. [20] secured VANET cloud systems using ML classifiers for predictive analytics on vehicular network flow. Mahato et al. [21] applied a chaotic grey wolf optimizer with KNN to detect attacks in healthcare and IoMT networks. Recent studies have introduced hybrid or scalable

strategies. Fan et al. [22] utilized an improved distributed tensor train model with LightGBM to detect attacks across heterogeneous networks. Belachew et al. [23] deployed SDN-enabled ML models at the edge for latency-aware mitigation policies. Pradeep and Shukla [24] combined feature extraction via a predefined-mud ring algorithm with deep architectures, including CAE and GRU. Dilshad et al. [25] adopted federated learning and Gini-based feature selection for DDoS detection in the Internet of Vehicles. Hosain and Çakmak [26] developed an explainable intrusion detection system for IoMT using recursive feature elimination and XGBoost, integrated with SHAP and LIME for interpretability. Although ML-based approaches show promise, they are often constrained by manual feature engineering, limited adaptability to evolving attack patterns, and performance degradation under data imbalance. These limitations motivate the adoption of deep learning and reinforcement learning techniques for more robust and generalizable DDoS detection frameworks.

B. Deep Learning

DL has significantly enhanced network security by providing advanced solutions for DDoS attack detection. Traditional methods based on specialized networks such as CNN and long short-term memory have effectively addressed recognizing DDoS attacks in various network environments.

Benzaïd et al. [27] proposed a defense framework that uses a context-gated GRU (CG-GRU) model to learn spatiotemporal patterns and applies transfer learning to detect economic denial of sustainability (EDoS) attacks in flexible beyond fifth-generation (B5G) network slicing environments. Duan et al. [28] proposed a cyber anomaly detection framework based on a continuous temporal graph neural network. This network models dynamic interactions and applies spatio-temporal message aggregation to handle irregular and low-frequency network behaviors. Shieh et al. [29] introduced a CNN-based method combined with reciprocal points learning and open-set recognition to detect both known and unknown DDoS attacks. The lightweight model minimizes training parameters, offering an efficient and flexible solution for cybersecurity. Anley et al. [30] developed a DDoS detection method using CNN, adaptive architectures, and transfer learning. This approach enhances pattern recognition in traffic data and improves generalization across datasets, facilitating the distinction between legitimate and malicious behaviors in network infrastructures. Farzaneh et al. [31] applied deep transfer learning to improve DDoS detection in 5G networks. They utilized models such as bidirectional LSTM (BiLSTM), CNN, residual networks (ResNet), and Inception to transfer knowledge from large datasets to smaller, annotated ones, which helps overcome challenges related to limited data availability. Ali et al. [32] proposed a vision transformer (ViT)-based system that transforms DDoS attack traffic into images and applies transformer-based deep learning for detecting multi-vector attacks during firmware over-the-air (OTA) updates. Ullah et al. [33] introduced an explainable hybrid model that combines CNN for extracting spatial features and LSTM for learning temporal patterns, with SHAP used to

interpret decisions in DDoS detection for SDN-IoT healthcare settings. Doménech et al. [34] developed a medical intrusion detection system for the IOMT to counter DDoS attacks, employing ML models with feature optimization and domain-specific preprocessing techniques.

C. RL and DRL

Several studies have explored traditional reinforcement learning (RL) methods, such as Q-learning and adaptive rate control, to detect and mitigate DDoS attacks in dynamic IoT environments. Kumar and Singh [35] developed a message-driven RL (MD-RL) framework for IoT edge security, utilizing state-action policies for dynamic detection and response to DDoS threats. Said et al. [36] proposed a quantum entropy Q-learning (QEQ) framework that extends classical RL by integrating quantum processing and entropy-driven exploration, enhancing DDoS mitigation in smart grid environments. Jayakrishna and Prasanth [37] introduced an RL-based adaptive rate limiting (RL-ARL) technique that adjusts rate thresholds in vehicular ad hoc networks (VANETs) based on traffic severity to counter DDoS attacks.

Recent research has increasingly adopted deep reinforcement learning (DRL) techniques, such as deep Q-networks and policy optimization, to improve DDoS detection in complex systems. Roy et al. [5] combined a multi-step deep Q-learning network (MSDQN) with a deep learning network (DLN) to secure healthcare IoT systems. The DLN verifies IoT device identities, while MSDQN detects and mitigates malware and DDoS threats during data transmission. Gai et al. [38] introduced a defense mechanism using deep Q-learning for moving target defense in consortium blockchains, dynamically adjusting attack surfaces based on learned policies. Paidipati et al. [39] proposed a multilayer ensemble DRL (MEDRL) approach, incorporating metaheuristic tuning via an improved grasshopper optimization algorithm (IGO) to detect and mitigate DDoS attacks in cloud-based software-defined networks (SDNs). Shaikh et al. [40] developed a model for IoMT healthcare networks that integrates CNN, LSTM, and reinforcement learning, leveraging a deep Q-network and PPO for robust DDoS countermeasures.

D. Limitations

Deep learning plays a key role in feature extraction for DDoS attack detection but faces several limitations. It often requires extensive preprocessing to identify important features. DDoS datasets are typically imbalanced, with normal traffic dominating, which can lead to biased predictions. Models may also struggle to generalize to new or evolving attack patterns. In addition, hyperparameter tuning is complex and time-consuming, affecting both performance and scalability. While RL and DRL models offer dynamic adaptability for feature selection and classification in imbalanced settings, they commonly suffer from sample inefficiency, limiting their effectiveness. To overcome these issues, this study introduces an off-policy PPO algorithm for feature selection and imbalanced classification, an improved GAN for data augmentation, and BOHB for efficient hyperparameter optimization.

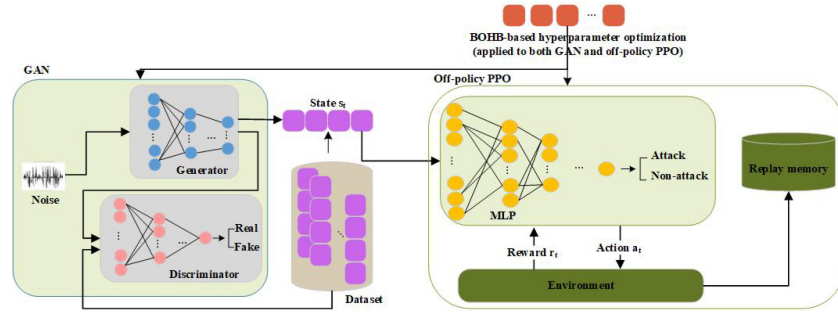


Fig. 1. Architecture of the suggested DDoS detection model using off-policy PPO for feature selection and imbalanced classification, an improved GAN method for online data augmentation, and the BOHB method for hyperparameter optimization.

Algorithm 1 Pseudo-Code for Online Data Augmentation Algorithm

Input:

p : Threshold probability used to decide whether to retain the original sample

p_I : Computed probability for retaining sample I : Collection of mini-batches from the original dataset

N : The overall number of mini-batches

B_i : A specific mini-batch from B : An individual sample within mini-batch B_i

y_I : The label associated with sample I : A newly generated sample produced by the GAN with the same label y_I

B_{aug} : The final dataset composed of augmented mini-batches

Output: Expanded dataset B_{aug}

Begin with initializing initial batches $B = \{B_1, B_2, B_3, \dots, B_N\}$

Assign B_{aug} as an empty collection

for each B_i in B :

 Compute p for each sample in B_i

for each I in B_i :

if $p_I < p$

 Move to the next sample (omit any modifications)

else:

 Determine y_I

 Generate I_N

 Substitute sample I with I_N

 Append the possibly modified B_i to B_{aug}

return B_{aug}

gradients from the most influential mini-batch elements during updates, enhancing both stability and sample diversity. The update process is as follows:

$$L_G = \theta_G - \alpha_G \sum_{Z_i \in Z \setminus \text{top}_k\{D(Z)\}} \nabla_{\theta_G} V(D(z_i), G). \quad (2)$$

Here, $D(Z)$ represents the discriminator's response to each sample in the mini-batch Z . θ_G is the generator's parameter set, and α_G is the learning rate. The expression $\sum_{Z_i \in Z \setminus \text{top}_k\{D(Z)\}} \nabla_{\theta_G} V(D(z_i), G)$ computes gradients only from mini-batch samples excluding the top-k highest-scoring ones as determined by the discriminator. These top-k samples strongly align with the target distribution and heavily influence model updates. To prevent overfitting to such dominant features, our method excludes their gradients, promoting better generalization and sample diversity. This strategy also mitigates mode collapse and stabilizes training by encouraging the generator to explore a wider output space, leading to more diverse and robust results.

B. Feature Engineering

Consider D that includes pairs (x, y) , where x is a vector containing n input features $F = \{f_1, \dots, f_n\}$, and y represents the outcome. Each iteration involves randomly selecting a pair from D , and the agent determines which features to utilize. In our RL framework, the state space S comprises the data instance x , its outcome y , and the chosen features F . The possible actions in our model are part of the set A , which entails selecting a feature A_f or making a prediction A_c . Opting for a feature from A_f adds to the feature set F , thus modifying the prediction space. The reward for selecting a feature is given by $\partial * \text{SHAP}(f_i)$, where $\text{SHAP}(f_i)$ denotes the SHAP value of the feature f_i , and ∂ is a balancing hyperparameter. Actions that lead to predictions shift the model to a terminal state, offering rewards or penalties of ± 1 for the majority class and $\pm \mu$ for the minority class. So, $r: S \times A \rightarrow R$ is mathematically defined to encapsulate these dynamics:

$$r_t = \begin{cases} \partial * \text{SHAP}(f_i) & \text{if } a_t \in A_f, a_t = f_i \\ +1 & \text{if } a_t \in A_c, s_t \in C_O, \hat{y} = y \\ -1 & \text{if } a_t \in A_c, s_t \in C_O, \hat{y} \neq y \\ +\mu & \text{if } a_t \in A_c, s_t \in C_N, \hat{y} = y \\ -\mu & \text{if } a_t \in A_c, s_t \in C_N, \hat{y} \neq y \end{cases} \quad (3)$$

III. THE PROPOSED MODEL

A. Online Data Augmentation

In this paper, we propose a GAN-based online data augmentation method that enhances sample diversity by selectively omitting gradients from key batch elements. As shown in Algorithm 1, the generator and discriminator are trained in a two-player minimax game, where the generator aims to produce realistic synthetic data while the discriminator learns to distinguish real from fake samples.

$$\min_G \min_D V(D, G) = E_{x \sim p_{\text{data}}(x)} [\log(D(x))] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (1)$$

Here, x and z represent the input data and random noise, respectively, while G and D denote the generator and discriminator. $p_{\text{data}}(x)$ and $p_z(z)$ are the probability distributions of input data and noise, and E and $\log$ denote expectation and logarithm. In this study, the generator uses a novel gradient exclusion strategy to improve generalization and diversity by removing dominant training signals. It selectively excludes

where C_O and C_N represent the minority and majority classes. The reward mechanism emphasizes accurate recognition across all classes, especially minority ones, to ensure fairness. Integrating SHAP values into the reward system enhances transparency and interpretability in DDoS detection for healthcare. SHAP offers insight into feature contributions, supporting trustworthy decision-making and helping mitigate class bias. This improves detection accuracy while ensuring fairness, providing robust protection for sensitive healthcare systems against diverse threats.

SHAP values are based on the concept of the Shapley value from cooperative game theory. The formula for the SHAP value ϕ_i for a feature i is:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} (v(S \cup \{i\}) - v(S)) \quad (4)$$

where F denotes the full set of features, and S represents any subset of F that does not include feature i . The notation $|S|$ refers to the number of elements in subset S , while $|F|$ represents the total number of features. The value $v(S)$ indicates the output of the model when only the features in S are used, and $v(S \cup \{i\})$ represents the prediction of the model when feature i is added to subset S . Overall, SHAP measures how much adding feature i to subset S changes the output of the model, averaged over all possible subsets.

The transition function of the model operates as described below:

$$t((x, y, \mathbb{F}), a) = \begin{cases} T & \text{if } a \in A_c \\ (x, y, \mathbb{F} \cup a) & \text{if } a \in A_f \end{cases} \quad (5)$$

where T denotes the terminal state. When an action adds a new feature, it is included in the set F . The episode concludes when the agent opts to make a prediction. It is crucial to note that our RL model functions within a discrete-time framework. The time variable t in s_t and s_{t+1} marks when decisions to incorporate features are made. In reinforcement learning, each decision involves selecting a feature that can enhance future prediction accuracy, influencing subsequent states and rewards.

Algorithm 2 outlines the learning strategy involving dataset shuffling, policy-based action selection, model execution, and feedback-driven updates using the off-policy PPO algorithm.

1) Training: To enhance policy training, we adopt an off-policy PPO algorithm. This section begins with an overview of trust region policy optimization (TRPO), followed by PPO as its foundation, and concludes with the proposed off-policy PPO approach.

TRPO: To optimize policy performance, we first review Trust Region Policy Optimization (TRPO), which maximizes expected discounted rewards from the initial state s_t :

$$\mathbb{E}_{(s_0, a_0), \dots} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \quad (6)$$

To ensure stable policy updates, TRPO maximizes a surrogate objective under a Kullback–Leibler (KL) divergence constraint:

$$\text{Maximum}_{\pi_{\theta_{i+1}}} \mathbb{E}_{s \sim \rho_{\pi_{\theta_i}}}, a \in \pi_{\pi_i} \left[\frac{\pi_{\theta_{i+1}}(a|s)}{\pi_{\theta_i}(a|s)} A_{\pi_i}(s, a) \right] \quad (7)$$

Algorithm 2 Pseudo-Code for Training the Proposed Model

Input:

$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_T, y_T)\}$: Training dataset with T labeled samples

α : Learning rate used for policy update

E : Total number of training epochs

ϵ : Exploration probability (for selecting random actions)

θ_i : Parameters of the current policy (old policy)

θ_{i+1} : Parameters of the updated policy (new policy)

ϕ : Parameters of the value function (critic)

s_t : State at time t ; a_t : Action taken at time t ; r_t : Reward received at time t

$A(s_t, a_t)$: Advantage estimation at time t for state-action pair

$\mu(\cdot|s_t)$: Behavior policy used to sample action a_t at state s_t : The target policy to be optimized

B : Replay buffer to store transitions

Q : Action-value function approximator (initialized with random weights)

Initialize θ_{i+1} , θ_i , ϕ , A , B , and Q

for $e = 1$ to E :

Randomize order of D

for $t = 1$ to T :

Select a_t with probability ϵ

Set s_1

if action a_t is feature-related:

Perform action a_t

Calculate r_t using Equation (3)

Determine s_{t+1} from D

Store transition $(s_t, a_t, r_t, s_{t+1}, \mu(\cdot|s_t))$ in B

Compute $A(s_t, a_t)$ using value function parameters ϕ

Update π optimize the objective function via Equation (19):

$$\theta_{i+1} \leftarrow \theta_i + \alpha \nabla_{\theta} L_{\text{Off-Policy}}^{\text{CLIP}} \text{PPO}(\pi)$$

if a_t is subject-related:

Perform a_t

Calculate r_t using Equation (3)

Determine s_{t+1} from D

Store transition $(s_t, a_t, r_t, s_{t+1}, \mu(\cdot|s_t))$ in B

subject to:

$$\mathbb{E}_{s \sim \rho_{\pi_i}} [\text{Distance}_{\text{KL}}(\pi_i(\cdot|s) || \pi_{i+1}(\cdot|s))] \leq \vartheta \quad (8)$$

where π_{θ_i} and $\pi_{\theta_{i+1}}$ represent the current and subsequent policies, respectively. ϑ establishes the threshold for allowable divergence. $\text{Distance}_{\text{KL}}$ measures the KL divergence, showing the extent of deviation between the policies π_{θ_i} and $\pi_{\theta_{i+1}}$ at state s . The distribution $\rho_{\pi_{\theta_i}}$ details the probability distribution of states under the old policy, beginning from the initial state s_0 and adhering to π_i , calculated as $\rho_{\pi_{\text{old}}}(s) = \sum_{t=0}^{\infty} \gamma^t P(s_t = s | s_0, \pi_{\text{old}})$, with P being the transition probability. In the absence of this divergence limit, improving the surrogate objective could lead to major alterations in the policy.

PPO: Proximal Policy Optimization (PPO) avoids abrupt policy updates by optimizing a clipped surrogate objective:

$$\mathbb{E}_{s \sim \rho_{\pi_i}, a \sim \pi_i} \left[\text{minimum} \left(\frac{\pi_{i+1}(a|s)}{\pi_i(a|s)} A_{\pi_i}(s, a), \text{clip} \left(\frac{\pi_{i+1}(a|s)}{\pi_i(a|s)}, 1 - \epsilon, 1 + \epsilon \right) A_{\pi_i}(s, a) \right) \right] \quad (9)$$

where ϵ is a small constant to limit policy deviation and encourage stability. $A_{\pi_i}(s, a)$ represents the advantage function, quantifying the extra benefit of taking action a in state s .

s according to policy π_i . This function evaluates if an action performs better or worse than what is typically expected by that policy. In this context, clipping restricts excessive training updates, promoting more consistent progress. This method is further detailed below:

$$\text{clip}(p, x, y) = \text{maximum}(x, \text{minimum}(y, p)) \quad (10)$$

where p represents the variable under modification, with x and y setting the minimum and maximum boundaries for the allowable range. It is crucial to recognize that the clipped surrogate objective (Equation (9)) aims to curtail excessively large updates to the policy by penalizing deviations that cause the ratio $\frac{\pi_{i+1}(a|s)}{\pi_i(a|s)}$ to diverge significantly from 1. Nonetheless, a primary issue with PPO is its dependence on on-policy data, which escalates the sample complexity due to the restricted utilization of off-policy data, requiring more frequent interactions between the agent and the environment.

Off-policy PPO: Off-policy PPO improves data efficiency by learning from past experiences, including historical DDoS incidents. Unlike on-policy PPO, it uses off-policy data to recognize evolving attack patterns and adapt to changes.

$$\text{maximum}_{\pi} \mathbb{E}_{s \sim \rho_{\mu}, a \in \mu} \left[\frac{\pi_{i+1}(a|s)}{\mu(a|s)} A_{\pi_i}(s, a) \right] \quad (11)$$

with:

$$\overline{\text{Distance}}_{\text{KL}}^{\rho_{\mu}, \text{sqrt}}(\mu, \pi_i) \overline{D}_{\text{KL}}^{\rho_{\mu}, \text{sqrt}}(\pi_i, \pi_{i+1}) + \overline{D}_{\text{KL}}^{\rho_{\mu}}(\pi_i, \pi_{i+1}) \leq \vartheta \quad (12)$$

$$\rho_{\mu}(s) = \sum_{t=0}^{\infty} \gamma^t P(s_t = s | s_0, \mu) \quad (13)$$

$$\overline{\text{Distance}}_{\text{KL}}^{\rho_{\mu}}(\pi_{\text{old}}, \pi) = \mathbb{E}_{s \sim \rho_{\mu}} [\text{Distance}_{\text{KL}}(\pi_i(\cdot|s) || \pi_{i+1}(\cdot|s))] \quad (14)$$

$$\overline{\text{Distance}}_{\text{KL}}^{\rho_{\mu}, \text{sqrt}}(\mu, \pi_{\text{old}}) = \mathbb{E}_{s \sim \rho_{\mu}} \left[\sqrt{\text{Distance}_{\text{KL}}(\mu(\cdot|s) || \pi_i(\cdot|s))} \right] \quad (15)$$

$$\overline{\text{Distance}}_{\text{KL}}^{\rho_{\mu}, \text{sqrt}}(\pi_i, \pi_{i+1}) = \mathbb{E}_{s \sim \rho_{\mu}} \left[\sqrt{\text{Distance}_{\text{KL}}(\pi_i(\cdot|s) || \pi_{i+1}(\cdot|s))} \right] \quad (16)$$

where μ denotes the behaviour policy, showing the typical decision-making strategy the agent uses in the environment. Absent the constraints (Equation (12)), efforts to improve the surrogate function using off-policy data (Equation (10)) could result in significant alterations to the policy. To mitigate this potential problem, we utilize the PPO clipping method, modifying the surrogate objective as follows:

$$L_{\mu}(\pi_{i+1}) = \mathbb{E}_{s \sim \rho_{\mu}, a \in \mu} \left[\frac{\pi(a|s)}{\mu(a|s)} A_{\pi_{\text{old}}}(s, a) \right] \quad (17)$$

We can specify the clipped surrogate objective for off-policy data as follows:

$$\begin{aligned} \bar{L}_{\mu}(\pi_{i+1}) &= \mathbb{E}_{s \sim \rho_{\mu}, a \in \mu} \left[\text{minimum} \left(\frac{\pi_{i+1}(a|s)}{\mu(a|s)} A_{\pi_i}(s, a), \right. \right. \\ &\quad \left. \left. \text{clip} \left(\frac{\pi_{i+1}(a|s)}{\mu(a|s)}, 1 - \varepsilon, 1 + \varepsilon \right) A_{\pi_i}(s, a) \right) \right] \quad (18) \end{aligned}$$

TABLE I
HYPERPARAMETERS AND THEIR RANGES FOR OPTIMIZATION IN THE MODEL

Hyperparameter	Ranges	Best value		
		Generator	Discriminator	MLP
Batch size	[8, 512]	KDDcup99 (64), ISCX-UNB (58), and CICDDoS (43)	KDDcup99 (38), ISCX-UNB (56), and CICDDoS (32)	KDDcup99 (42), ISCX-UNB (65), and CICDDoS (40)
Epoch	[32, 1024]	KDDcup99 (132), ISCX-UNB (165), and CICDDoS (168)	KDDcup99 (98), ISCX-UNB (102), and CICDDoS (136)	KDDcup99 (156), ISCX-UNB (215), and CICDDoS (174)
Learning rate	[0, 1]	KDDcup99 (0.002), ISCX-UNB (0.003), and CICDDoS (0.005)	KDDcup99 (0.0003), ISCX-UNB (0.002), and CICDDoS (0.001)	KDDcup99 (0.002), ISCX-UNB (0.005), and CICDDoS (0.004)
Activation function	{Leaky ReLU, ReLU, Tanh, Linear, Sigmoid}	KDDcup99 (ReLU), ISCX-UNB (ReLU), and CICDDoS (ReLU)	KDDcup99 (ReLU), ISCX-UNB (ReLU), and CICDDoS (ReLU)	KDDcup99 (ReLU), ISCX-UNB (ReLU), and CICDDoS (ReLU)
Dropout rate	[0, 1]	KDDcup99 (0.6), ISCX-UNB (0.4), and CICDDoS (0.5)	KDDcup99 (0.4), ISCX-UNB (0.6), and CICDDoS (0.6)	KDDcup99 (0.3), ISCX-UNB (0.7), and CICDDoS (0.5)
Layers in MLP	[1, 8]	KDDcup99 (5), ISCX-UNB (3), and CICDDoS (4)	KDDcup99 (4), ISCX-UNB (3), and CICDDoS (5)	KDDcup99 (7), ISCX-UNB (5), and CICDDoS (6)
Hidden size	[8, 1024]	KDDcup99 (69), ISCX-UNB (52), and CICDDoS (51)	KDDcup99 (73), ISCX-UNB (68), and CICDDoS (53)	KDDcup99 (59), ISCX-UNB (45), and CICDDoS (49)
Batch size	[8, 512]	KDDcup99 (64), ISCX-UNB (58), and CICDDoS (62)	KDDcup99 (38), ISCX-UNB (56), and CICDDoS (48)	KDDcup99 (42), ISCX-UNB (65), and CICDDoS (52)

The value $\frac{\pi_{i+1}(a|s)}{\mu(a|s)}$ often exceeds the boundaries of $1 - \varepsilon$ to $1 + \varepsilon$. As a result, the policy $\pi_{i+1}(a|s)$ typically stays consistent when optimizing the clipped surrogate objective. To counteract this rigidity, the boundaries of the clipped objective, ranging from $1 - \varepsilon$ to $1 + \varepsilon$, are adjusted in Equation (18) by incorporating a corrective factor $\frac{\pi_{i+1}(a|s)}{\mu(a|s)}$:

$$\begin{aligned} \mathbb{E}_{s \sim \rho_{\mu}, a \in \mu} &\left[\text{minimum} \left[\frac{\pi_{i+1}(a|s)}{\mu(a|s)} A_{\pi_i}(s, a), \right. \right. \\ &\quad \left. \left. \text{clip} \left(\frac{\pi_{i+1}(a|s)}{\mu(a|s)}, \frac{\pi_i(a|s)}{\mu(a|s)} (1 - \varepsilon), \frac{\pi_i(a|s)}{\mu(a|s)} (1 + \varepsilon) \right) A_{\pi_i}(s, a) \right] \right] \quad (19) \end{aligned}$$

C. Hyperparameter Optimization

To address the high cost of evaluations, BOHB integrates Hyperband to speed up optimization through adaptive resource allocation and early stopping. It tests multiple configurations, focuses resources on the most promising, and eliminates weaker ones, lowering computation time and cost. By combining Bayesian search with Hyperband's efficiency, BOHB enables fast and accurate hyperparameter tuning in dynamic environments like healthcare cybersecurity. This helps ensure strong model performance across diverse attack scenarios and improves the protection of sensitive healthcare data. Table I lists the hyperparameters optimized using BOHB.

IV. EXPERIMENTS

A. Dataset

To evaluate the proposed DDoS detection model, datasets such as the KDDcup99 [41], the information security center of excellence at the University of New Brunswick

TABLE II
PERFORMANCE COMPARISON OF THE PROPOSED MODEL AGAINST ML, DL AND ABLATED MODELS USING THE KDDCUP99 DATASET

Model	Accuracy	Precision	Recall	F-measure	G-means	MSE	Runtime	GPU
DPTCM-KNN [41]	70.103 ± 0.004	68.172 ± 0.047	66.534 ± 0.024	67.343 ± 0.037	68.172 ± 0.064	9.712 ± 0.066	2956	17.5
ERF-KMC [42]	70.607 ± 0.019	71.686 ± 0.056	68.121 ± 0.023	69.858 ± 0.093	71.686 ± 0.027	9.717 ± 0.060	2856	18.3
FSSVM [1]	71.635 ± 0.087	71.318 ± 0.052	67.765 ± 0.045	69.496 ± 0.071	71.318 ± 0.032	8.730 ± 0.090	2145	19.2
ML-CFS [6]	72.500 ± 0.082	72.632 ± 0.086	68.995 ± 0.034	70.767 ± 0.056	72.632 ± 0.017	9.136 ± 0.012	1896	15.6
ML-EA [43]	73.134 ± 0.034	74.830 ± 0.047	69.222 ± 0.040	71.917 ± 0.097	74.830 ± 0.071	8.755 ± 0.062	2652	17.2
IDAD [23]	74.339 ± 0.082	74.729 ± 0.051	70.985 ± 0.078	72.809 ± 0.002	74.729 ± 0.091	8.769 ± 0.072	2547	19.3
SDN-Edge-IoT [24]	74.958 ± 0.020	75.324 ± 0.051	71.581 ± 0.023	73.405 ± 0.011	75.324 ± 0.009	7.777 ± 0.082	2136	18.4
CNN-RPL [29]	77.729 ± 0.007	77.910 ± 0.070	76.315 ± 0.079	77.104 ± 0.004	77.910 ± 0.043	7.789 ± 0.023	2963	22.7
CNN-BiLSTM [44]	78.886 ± 0.030	78.157 ± 0.036	78.438 ± 0.079	78.297 ± 0.033	78.157 ± 0.013	7.804 ± 0.008	3142	25.6
ATL [30]	79.673 ± 0.003	80.081 ± 0.099	76.346 ± 0.095	78.169 ± 0.046	80.081 ± 0.084	8.819 ± 0.009	3462	24.3
LSTM-CNN [45]	80.936 ± 0.031	80.880 ± 0.058	77.194 ± 0.098	78.994 ± 0.018	80.880 ± 0.076	5.827 ± 0.073	3569	26.0
MDDCC [46]	81.956 ± 0.086	82.148 ± 0.017	80.589 ± 0.052	81.361 ± 0.028	82.148 ± 0.086	5.834 ± 0.006	3863	25.7
DTL-5G [31]	82.704 ± 0.035	83.233 ± 0.048	81.546 ± 0.031	82.381 ± 0.096	83.233 ± 0.058	4.840 ± 0.029	3952	26.3
FCNN [3]	83.195 ± 0.054	84.373 ± 0.100	82.810 ± 0.033	83.584 ± 0.013	84.373 ± 0.052	4.852 ± 0.065	4012	28.2
mGRU [47]	84.524 ± 0.095	85.870 ± 0.088	82.336 ± 0.100	84.066 ± 0.012	85.870 ± 0.046	4.861 ± 0.066	3896	20.7
DDoSViT [32]	85.940 ± 0.058	84.940 ± 0.038	83.924 ± 0.037	84.429 ± 0.082	84.940 ± 0.062	3.745 ± 0.041	3368	19.7
MD-RL [21]	75.404 ± 0.065	74.899 ± 0.079	74.081 ± 0.089	74.488 ± 0.027	74.899 ± 0.048	6.633 ± 0.031	3412	21.2
Q-learning [34]	84.740 ± 0.010	83.787 ± 0.019	82.819 ± 0.021	83.300 ± 0.003	83.787 ± 0.072	4.734 ± 0.092	3045	23.5
MEDRL [35]	85.240 ± 0.058	85.946 ± 0.023	82.965 ± 0.042	84.429 ± 0.082	85.946 ± 0.062	4.745 ± 0.041	3628	22.7
RL-ARL [36]	86.900 ± 0.099	85.682 ± 0.014	85.140 ± 0.027	85.410 ± 0.008	85.682 ± 0.025	3.762 ± 0.018	3265	21.4
Proposed w/o DA	87.797 ± 0.098	87.993 ± 0.047	84.484 ± 0.035	86.203 ± 0.050	87.993 ± 0.029	2.879 ± 0.010	N/A	N/A
Proposed w/o FS	82.656 ± 0.035	79.223 ± 0.059	81.631 ± 0.025	80.409 ± 0.096	79.223 ± 0.068	3.895 ± 0.087	N/A	N/A
Proposed w/o off-policy PPO	85.556 ± 0.003	84.404 ± 0.100	84.944 ± 0.088	84.673 ± 0.064	84.404 ± 0.012	2.909 ± 0.006	N/A	N/A
Proposed w/o HO	87.476 ± 0.003	87.925 ± 0.010	84.630 ± 0.088	86.246 ± 0.075	87.925 ± 0.012	2.927 ± 0.096	N/A	N/A
Proposed using only PPO + SHAP	88.476 ± 0.024	87.978 ± 0.026	86.292 ± 0.088	87.127 ± 0.035	87.978 ± 0.018	1.868 ± 0.082	N/A	N/A
Proposed model	90.498 ± 0.095	90.983 ± 0.011	87.651 ± 0.031	89.286 ± 0.096	90.983 ± 0.047	1.142 ± 0.027	2726	19.6

(ISCX-UNB) [42], the Canadian Institute for Cybersecurity DDoS (CICDDoS) [43], and the Defense Advanced Research Projects Agency (DARPA) [44] were employed, each contributing values that are pertinent to healthcare industries.

B. Evaluation Metrics

These following metrics used to evaluate the proposed method:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total TP}} \quad (20)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (21)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (22)$$

$$F - \text{Measure} = 2 \times \frac{\text{Precision} + \text{Recall}}{\text{Precision} + \text{Recall}} \quad (23)$$

$$G - \text{means} = \sqrt{\text{Recall} \times \text{Specificity}} \quad (24)$$

with

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (25)$$

C. Experiment Setting and Main Evaluation Results

The study was conducted on a 64-bit Windows system with an Intel i7 processor, 32 GB RAM, and an NVIDIA RTX 3090 GPU, supporting large-scale data and complex computations. Python 3.8 was used with PyTorch for deep learning tasks, including the GAN and off-policy PPO algorithm. Scikit-learn handled traditional machine learning and data preprocessing. Hyperparameter tuning was performed using BOHB via the HpBandSter library. SHAP was used to interpret model predictions by identifying key features.

In the evaluation phase, our model was extensively compared with seven ML models, including DPTCM-KNN [45], ERF-KMC [46], FSSVM [1], ML-CFS [6], ML-EA [47],

IDAD [22], and SDN-Edge-IoT [23]; nine DL models, including CNN-RPL [29], CNN-BiLSTM [48], ATL [30], LSTM-CNN [49], MDDCC [50], DTL-5G [31], FCNN [3], mGRU [51], and DDoSViT [32]; and four RL models, including MD-RL [35], Q-learning [38], MEDRL [39], and RL-ARL [37]. Additionally, we evaluated the impact of removing critical elements such as data augmentation (DA), feature selection (FS), off-policy PPO, and hyperparameter optimization (HO) from our model to understand their contributions to its overall performance. Tables II, III, and IV show the result details for the KDDcup99, ISCX-UNB, and CICDDoS datasets.

Component analysis shows that removing data augmentation, feature selection, off-policy PPO, or hyperparameter optimization leads to noticeable performance declines, highlighting the importance of these elements. Although simplified versions such as PPO with SHAP provide weaker supervision, they remain applicable for resource-limited scenarios. A two-tailed paired t-test was conducted to verify effectiveness, with p-values consistently below 0.05 (accuracy p = 0.012, F-measure p = 0.015, G-means p = 0.009), indicating strong statistical significance. Confidence intervals further confirm the observed improvements, supporting the model's reliability for DDoS detection in healthcare settings.

Tables II, III, and IV present a comparison of computational efficiency among state-of-the-art DDoS detection models based on runtime and GPU usage across the KDDcup99, ISCX UNB, and CICDDoS datasets. The proposed model demonstrates improved efficiency, requiring 2726 seconds and 19.6 GB of GPU memory on KDDcup99 and 2587 seconds with 18.4 GB on ISCX UNB. Compared to the DTL 5G model, which uses 3696 seconds and 25.9 GB, the proposed model reduces runtime by 30.0% and GPU usage by 28.9%. On the CICDDoS dataset, it completes in 2659 seconds using 18.7 GB of GPU memory. These results confirm the model's ability to lower computational overhead.

TABLE III
PERFORMANCE COMPARISON OF THE PROPOSED MODEL AGAINST ML, DL AND ABLATED MODELS USING THE KDDCUP99 DATASET

Model	Accuracy	Precision	Recall	F-measure	G-means	MSE	Runtime	GPU
DPTCM-KNN [41]	71.511 ± 0.029	70.491 ± 0.012	70.036 ± 0.045	70.202 ± 0.030	71.929 ± 0.017	8.712 ± 0.066	2765	16.2
ERF-KMC [42]	72.790 ± 0.025	71.428 ± 0.067	70.926 ± 0.05	71.133 ± 0.093	72.807 ± 0.021	9.023 ± 0.060	2653	16.4
FSSVM [1]	72.434 ± 0.018	72.874 ± 0.032	71.806 ± 0.066	72.165 ± 0.071	73.868 ± 0.053	7.730 ± 0.090	2056	18.0
ML-CFS [6]	74.887 ± 0.046	73.985 ± 0.033	73.835 ± 0.022	73.969 ± 0.003	74.695 ± 0.021	7.742 ± 0.012	1963	14.3
ML-EA [43]	76.701 ± 0.075	73.383 ± 0.087	72.942 ± 0.058	72.290 ± 0.007	76.012 ± 0.091	7.755 ± 0.062	2425	16.7
IDAD [23]	77.957 ± 0.063	74.504 ± 0.099	73.314 ± 0.025	73.978 ± 0.078	75.682 ± 0.094	6.769 ± 0.072	2360	18.3
SDN-Edge-IoT [24]	78.615 ± 0.095	75.986 ± 0.018	74.952 ± 0.089	75.163 ± 0.027	77.799 ± 0.091	6.777 ± 0.082	2063	17.8
CNN-RPL [29]	79.215 ± 0.083	76.993 ± 0.049	75.982 ± 0.043	76.328 ± 0.040	78.008 ± 0.097	6.789 ± 0.023	2514	21.0
CNN-BiLSTM [44]	80.767 ± 0.011	77.994 ± 0.078	77.278 ± 0.079	77.484 ± 0.020	79.111 ± 0.049	6.804 ± 0.008	2974	24.3
ATL [30]	81.387 ± 0.041	78.907 ± 0.071	78.025 ± 0.017	78.140 ± 0.031	80.739 ± 0.000	5.819 ± 0.009	3214	22.1
LSTM-CNN [45]	83.983 ± 0.057	79.967 ± 0.064	78.714 ± 0.031	79.290 ± 0.062	80.942 ± 0.082	5.827 ± 0.073	3356	24.5
MDDCC [46]	83.409 ± 0.023	82.909 ± 0.022	81.148 ± 0.045	81.496 ± 0.067	82.097 ± 0.092	5.834 ± 0.006	3562	24.3
DTL-5G [31]	83.549 ± 0.088	82.631 ± 0.074	81.708 ± 0.06	82.048 ± 0.019	84.659 ± 0.032	6.840 ± 0.029	3696	25.9
FCNN [3]	85.684 ± 0.030	83.502 ± 0.087	82.747 ± 0.051	83.062 ± 0.018	85.729 ± 0.083	3.852 ± 0.065	3862	26.3
mGRU [47]	86.947 ± 0.002	84.516 ± 0.049	84.162 ± 0.064	84.261 ± 0.060	86.975 ± 0.023	3.861 ± 0.066	3620	19.5
DDoSViT [32]	86.127 ± 0.055	84.366 ± 0.021	83.097 ± 0.036	84.269 ± 0.069	85.059 ± 0.075	2.761 ± 0.091	3042	18.1
MD-RL [21]	77.544 ± 0.058	75.716 ± 0.07	75.882 ± 0.081	75.425 ± 0.061	76.181 ± 0.064	6.452 ± 0.024	3258	20.4
Q-learning [34]	84.696 ± 0.031	83.682 ± 0.045	83.515 ± 0.075	83.588 ± 0.037	84.398 ± 0.024	4.256 ± 0.068	2856	22.1
MEDRL [35]	86.127 ± 0.055	84.991 ± 0.054	84.065 ± 0.047	84.269 ± 0.069	85.059 ± 0.075	4.361 ± 0.091	3410	20.5
RL-ARL [36]	87.375 ± 0.053	86.343 ± 0.031	85.206 ± 0.032	85.596 ± 0.011	86.366 ± 0.087	2.771 ± 0.052	3042	20.8
Proposed w/o DA	89.549 ± 0.088	86.525 ± 0.072	85.722 ± 0.088	86.048 ± 0.019	86.659 ± 0.032	1.879 ± 0.010	N/A	N/A
Proposed w/o FS	86.684 ± 0.030	84.874 ± 0.058	83.839 ± 0.043	84.062 ± 0.018	82.729 ± 0.083	2.895 ± 0.087	N/A	N/A
Proposed w/o off-policy PPO	87.947 ± 0.002	83.885 ± 0.038	82.966 ± 0.016	83.261 ± 0.060	84.975 ± 0.023	1.909 ± 0.006	N/A	N/A
Proposed w/o HO	90.088 ± 0.034	89.545 ± 0.037	88.254 ± 0.024	88.814 ± 0.071	87.515 ± 0.053	1.927 ± 0.096	N/A	N/A
Proposed using only PPO + SHAP	90.552 ± 0.011	89.683 ± 0.015	88.652 ± 0.025	89.256 ± 0.052	88.265 ± 0.045	1.214 ± 0.035	N/A	N/A
Proposed model	93.186 ± 0.025	91.902 ± 0.019	90.869 ± 0.065	91.196 ± 0.055	91.896 ± 0.050	0.942 ± 0.027	2587	18.4

TABLE IV
PERFORMANCE COMPARISON OF THE PROPOSED MODEL AGAINST ML, DL AND ABLATED MODELS USING THE ISCX-UNB DATASET

Model	Accuracy	Precision	Recall	F-measure	G-means	MSE	Runtime	GPU
DPTCM-KNN [41]	73.572 ± 0.096	71.987 ± 0.091	71.293 ± 0.032	71.102 ± 0.079	70.48 ± 0.024	6.855 ± 0.099	2689	17.1
ERF-KMC [42]	74.845 ± 0.064	72.494 ± 0.093	71.133 ± 0.078	71.193 ± 0.076	72.496 ± 0.072	5.515 ± 0.057	2750	17.0
FSSVM [1]	75.447 ± 0.024	72.115 ± 0.028	71.466 ± 0.017	72.181 ± 0.067	73.455 ± 0.03	6.403 ± 0.077	1932	20.6
ML-CFS [6]	76.285 ± 0.088	73.029 ± 0.039	72.861 ± 0.025	73.767 ± 0.058	74.636 ± 0.031	1.048 ± 0.073	2006	16.9
ML-EA [43]	78.619 ± 0.074	74.809 ± 0.034	75.186 ± 0.083	73.951 ± 0.085	75.102 ± 0.077	5.283 ± 0.036	2326	16.3
IDAD [23]	78.229 ± 0.097	77.013 ± 0.042	76.093 ± 0.088	75.736 ± 0.027	77.09 ± 0.086	5.694 ± 0.083	2456	20.1
SDN-Edge-IoT [24]	81.129 ± 0.029	77.617 ± 0.059	78.213 ± 0.027	76.914 ± 0.063	78.834 ± 0.061	5.423 ± 0.092	1963	17.9
CNN-RPL [29]	79.545 ± 0.027	78.456 ± 0.082	78.833 ± 0.059	78.983 ± 0.011	78.535 ± 0.043	4.737 ± 0.055	2756	23.5
CNN-BiLSTM [44]	80.807 ± 0.057	79.061 ± 0.099	79.501 ± 0.091	79.496 ± 0.03	79.774 ± 0.032	5.327 ± 0.068	2863	24.9
ATL [30]	81.888 ± 0.036	80.576 ± 0.028	78.662 ± 0.02	79.839 ± 0.026	81.295 ± 0.045	4.14 ± 0.082	3352	23.6
LSTM-CNN [45]	82.908 ± 0.023	79.913 ± 0.083	79.828 ± 0.048	81.076 ± 0.045	82.234 ± 0.067	4.525 ± 0.04	3420	25.7
MDDCC [46]	83.341 ± 0.043	82.849 ± 0.076	82.244 ± 0.087	82.72 ± 0.022	83.217 ± 0.055	4.514 ± 0.018	3785	26.0
DTL-5G [31]	82.969 ± 0.081	83.803 ± 0.017	81.267 ± 0.056	82.409 ± 0.02	82.711 ± 0.054	3.937 ± 0.013	3605	25.4
FCNN [3]	84.087 ± 0.056	82.057 ± 0.02	81.514 ± 0.03	82.904 ± 0.089	83.265 ± 0.075	3.725 ± 0.059	3624	27.6
mGRU [47]	85.509 ± 0.014	85.567 ± 0.066	83.559 ± 0.04	84.084 ± 0.069	84.961 ± 0.012	4.381 ± 0.063	3541	19.9
DDoSViT [32]	86.796 ± 0.025	84.174 ± 0.016	84.727 ± 0.039	84.873 ± 0.06	85.349 ± 0.026	3.885 ± 0.013	3105	18.4
MD-RL [21]	86.336 ± 0.095	85.530 ± 0.039	85.41 ± 0.073	85.645 ± 0.032	87.302 ± 0.096	4.480 ± 0.042	2963	22.4
Q-learning [34]	87.361 ± 0.083	85.863 ± 0.067	84.157 ± 0.097	84.180 ± 0.091	86.628 ± 0.043	3.096 ± 0.057	2962	25.3
MEDRL [35]	86.816 ± 0.019	86.441 ± 0.052	85.915 ± 0.033	86.095 ± 0.067	85.095 ± 0.094	4.393 ± 0.029	3563	21.5
RL-ARL [36]	85.573 ± 0.05	83.516 ± 0.074	83.49 ± 0.037	83.639 ± 0.041	84.577 ± 0.097	3.105 ± 0.018	3105	20.7
Proposed w/o DA	86.067 ± 0.055	84.725 ± 0.061	83.003 ± 0.013	84.21 ± 0.091	85.397 ± 0.087	0.728 ± 0.058	N/A	N/A
Proposed w/o FS	81.796 ± 0.092	80.871 ± 0.054	79.884 ± 0.055	80.847 ± 0.08	81.352 ± 0.045	2.017 ± 0.067	N/A	N/A
Proposed w/o off-policy PPO	83.505 ± 0.07	80.423 ± 0.048	78.745 ± 0.035	80.345 ± 0.018	82.429 ± 0.039	2.202 ± 0.098	N/A	N/A
Proposed w/o HO	88.522 ± 0.057	84.796 ± 0.02	85.941 ± 0.032	84.721 ± 0.091	85.353 ± 0.06	1.591 ± 0.039	N/A	N/A
Proposed using only PPO + SHAP	90.01 ± 0.027	86.043 ± 0.067	85.3 ± 0.054	86.811 ± 0.011	88.111 ± 0.073	1.862 ± 0.034	N/A	N/A
Proposed model	92.403 ± 0.080	89.652 ± 0.056	89.970 ± 0.032	88.712 ± 0.070	90.285 ± 0.019	0.557 ± 0.017	2659	18.7

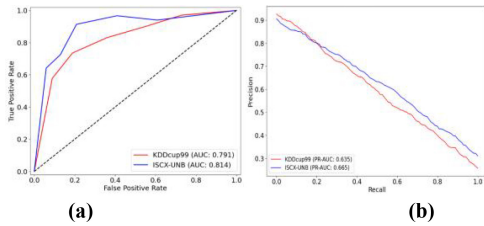


Fig. 2. Comparative a) ROC-AUC and b) PR-AUC curve analysis of the proposed model on the KDDcup99 and ISCX-UNB datasets.

Figure 2 shows the ROC and PR curves of the proposed model on the KDDcup99, ISCX-UNB, and CICDDOS datasets. The ROC AUCs are 0.791, 0.814, and 0.807, while

the PR AUCs are 0.635, 0.665, and 0.652, respectively, indicating strong discriminative ability and consistent precision. Figure 3 presents the training and validation loss curves over 300 epochs for the KDDcup99, ISCX-UNB, and CICDDOS datasets. Both curves steadily decline with a narrowing gap, indicating effective learning, minimal overfitting, and strong generalization.

Figure 4 presents the decision time distributions of the proposed model on the KDDcup99, ISCX UNB, and CICDDOS datasets, providing insight into its suitability for real-time systems. The histograms display consistent decision times across multiple runs, with distributions tightly centered around mean values of 62, 60, and 64 milliseconds, respectively. These results indicate a stable and efficient

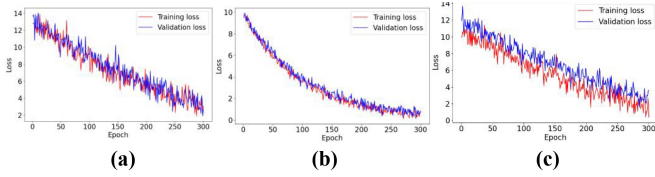


Fig. 3. Training and validation loss curves for the a) KDDcup99 and b) ISCX-UNB datasets in 300 epochs.

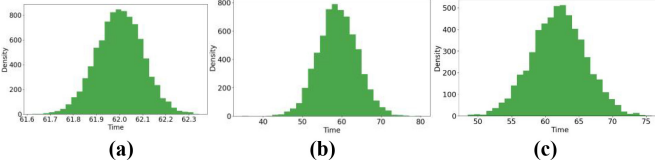


Fig. 4. Decision-making time distributions for the proposed model on the a) KDDcup99, b) ISCX-UNB, and c) CICDDOS datasets.

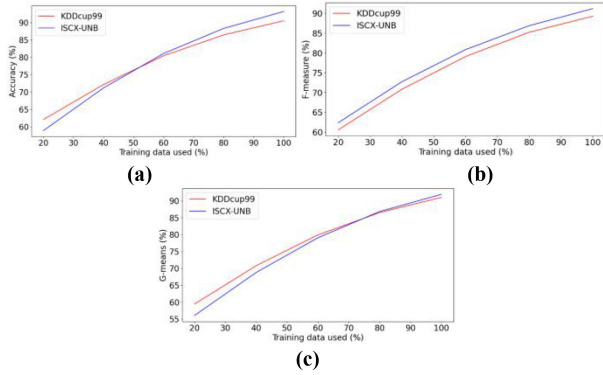


Fig. 5. Scalability of the proposed model demonstrated through improvements in a) accuracy, b) F-measure, and c) G-means with increasing training data from 20% to 100% on the KDDcup99, ISCX-UNB, and CICDDOS datasets.

decision-making process. Figure 5 illustrates the model's scalability as training data increases from 20% to 100% on the KDDcup99, ISCX-UNB, and CICDDOS datasets. Performance improves consistently, confirming the model's adaptability, robustness, and suitability for real-world scenarios with growing data volumes.

- 1) *Analysis of generalizability*: To validate the broad applicability of the proposed anomaly detection model, extensive evaluations were conducted on the DARPA dataset. As shown in Table XII, the model outperformed traditional machine learning and deep learning approaches, improving Accuracy by 6.54%, F-measure by 5.41%, and G-means by 5.35% compared to the next best model, mGRU. Component ablation analysis showed that removing any module, including data augmentation, feature selection, off-policy PPO, or hyperparameter optimization, resulted in noticeable drops in accuracy. These findings confirm the importance of each component in enhancing robustness, generalization, and detection performance.
- 2) *Analysis of off-policy PPO*: This section evaluates the stability of the Off-Policy PPO method by comparing its KL divergence and learning performance with PPO,

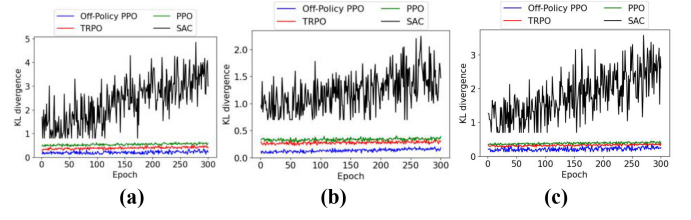


Fig. 6. KL divergence comparison across reinforcement learning algorithms for the a) KDDcup99, b) ISCX-UNB, and c) CICDDOS datasets in 300 epochs.

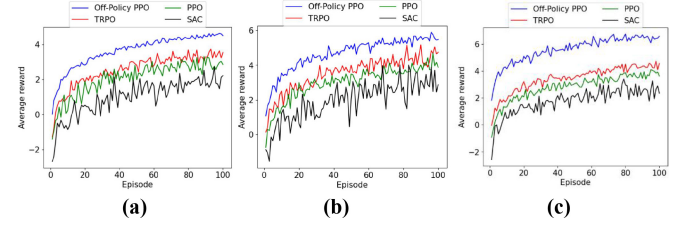


Fig. 7. Average reward comparison across reinforcement learning algorithms for the a) KDDcup99, b) ISCX-UNB, and c) CICDDOS datasets in 100 epochs.

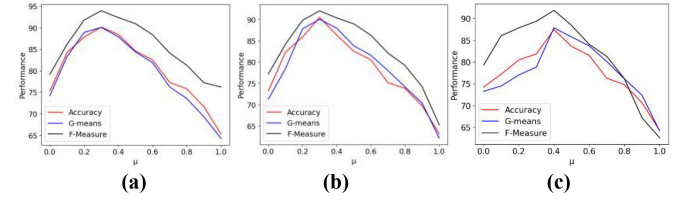


Fig. 8. Impact of varying reward modifier (μ) on the model performance.

TRPO, and soft actor-critic (SAC). Fig. 6 presents the KL divergence across policy updates on the KDDcup99, ISCX-UNB, and CICDDOS datasets. The KL divergence values of Off-Policy PPO remain consistently lower than those of SAC, indicating more stable policy updates and a reduced risk of disruptive shifts during training. Fig. 7 illustrates the learning curves through average rewards per episode. Off-policy PPO shows a smoother and faster increase in rewards compared to PPO, TRPO, and SAC, highlighting its ability to maintain consistent learning and adapt quickly.

Figure 8 shows the effect of varying μ (from 0 to 1 in 0.1 increments) on model performance for the KDDcup99, ISCX-UNB, and CICDDOS datasets. Performance peaks at $\mu = 0.3$, 0.3, and 0.4, respectively, indicating improved sensitivity to minority classes without sacrificing accuracy. This underscores the importance of tuning reward mechanisms in imbalanced settings and highlights the model's adaptability to diverse data distributions.

To enhance interpretability, SHAP was used to assess feature importance in DDoS detection, as shown in Fig. 9 for the KDDcup99, ISCX-UNB, and CICDDOS datasets. SHAP values highlight key features influencing model decisions. Table V further confirms that off-policy PPO effectively handles class imbalance, with minority class accuracy nearing that of majority classes. High F-measure and G-mean scores reflect

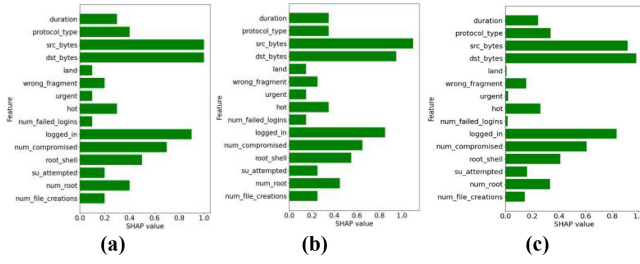


Fig. 9. Comparative analysis of feature importance using the SHAP explanations on the a) KDDcup99, b) ISCX-UNB, and c) CICDDOS datasets.

TABLE V
PERFORMANCE EVALUATION OF THE PROPOSED MODEL ON MAJORITY AND MINORITY CLASSES FOR THE KDDCUP99, ISCX-UNB, AND CICDDOS DATASETS

METRIC	KDDcup99		ISCX-UNB		CICDDOS	
	Majority	Minority	Majority	Minority	Majority	Minority
Accuracy	92.052 ± 0.072	85.756 ± 0.124	94.236 ± 0.032	88.563 ± 0.062	93.048 ± 0.082	87.852 ± 0.102
F-measure	91.547 ± 0.083	84.323 ± 0.152	92.722 ± 0.042	86.924 ± 0.093	91.692 ± 0.062	85.256 ± 0.023
G-means	91.823 ± 0.053	85.126 ± 0.103	93.512 ± 0.021	87.731 ± 0.071	92.658 ± 0.042	86.542 ± 0.045
MSE	0.963 ± 0.020	1.356 ± 0.032	0.902 ± 0.012	1.202 ± 0.025	0.924 ± 0.005	1.263 ± 0.045

TABLE VI
PERFORMANCE COMPARISON OF THE VARIOUS GANS FOR DATA AUGMENTATION USING THE KDDCUP99 DATASET

Model	FID	MMD	Wasserstein	KS
GAN	10.103 ± 0.064	14.343 ± 0.068	18.172 ± 0.049	0.785 ± 0.003
WGAN	9.727 ± 0.099	12.272 ± 0.025	17.057 ± 0.091	0.712 ± 0.057
SAGAN	9.023 ± 0.050	10.553 ± 0.026	16.362 ± 0.085	0.675 ± 0.067
PGGAN	8.150 ± 0.081	11.816 ± 0.023	12.661 ± 0.006	0.642 ± 0.060
BigGAN	6.859 ± 0.067	9.868 ± 0.002	13.684 ± 0.020	0.625 ± 0.062
StyleGAN	5.713 ± 0.009	7.056 ± 0.024	8.823 ± 0.088	0.612 ± 0.082
Proposed	1.022 ± 0.025	2.235 ± 0.052	3.124 ± 0.045	0.432 ± 0.001

balanced performance, and the slightly higher MSE for minority classes does not impact overall reliability. These results affirm the model's robustness and practical applicability.

3) *Analysis of data augmentation*: This section evaluates the proposed GAN for online data augmentation using Fréchet inception distance (FID), maximum mean discrepancy (MMD), Wasserstein distance, and the Kolmogorov–Smirnov (KS) test. These metrics measure statistical similarity between real and generated samples. Tables VI, VII, and X show results for the KDDcup99, ISCX-UNB, and CICDDOS datasets. On KDDcup99, the proposed GAN achieved lower scores with FID of 1.022, MMD of 2.235, Wasserstein distance of 3.124, and KS value of 0.432. Compared to StyleGAN, this corresponds to improvements of 82% in FID, 68% in MMD, 65% in Wasserstein distance, and 29% in KS value. Similar gains were observed on ISCX-UNB and CICDDOS, with metric reductions of up to 76%, 64%, 59%, and 32%, respectively. These improvements are attributed to the selective gradient exclusion strategy, which enhances sample diversity while maintaining balanced class representation.

Fig. 10 visualizes the alignment between the real and generated distributions for the original GAN and the improved GAN on the KDDcup99, ISCX-UNB, and CICDDOS datasets.

TABLE VII
PERFORMANCE COMPARISON OF THE VARIOUS GANS FOR DATA AUGMENTATION USING THE ISCX-UNB DATASET

Model	FID	MMD	Wasserstein	KS
GAN	8.511 ± 0.046	12.202 ± 0.043	16.929 ± 0.045	0.685 ± 0.028
WGAN	8.024 ± 0.056	11.324 ± 0.005	15.009 ± 0.053	0.681 ± 0.037
SAGAN	7.333 ± 0.066	10.964 ± 0.076	14.620 ± 0.081	0.626 ± 0.082
PGGAN	7.857 ± 0.014	9.297 ± 0.093	11.942 ± 0.001	0.593 ± 0.058
BigGAN	4.298 ± 0.000	8.093 ± 0.055	10.716 ± 0.014	0.582 ± 0.075
StyleGAN	4.073 ± 0.009	4.312 ± 0.007	4.983 ± 0.036	0.510 ± 0.026
Proposed	0.962 ± 0.051	1.535 ± 0.022	2.163 ± 0.037	0.356 ± 0.089

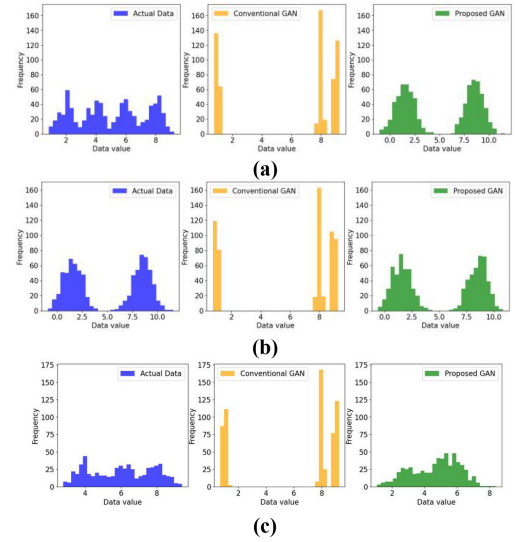


Fig. 10. Comparison of data distributions generated by the conventional GAN and the enhanced GAN for the a) KDDcup99, b) ISCX-UNB, and c) CICDDOS datasets.

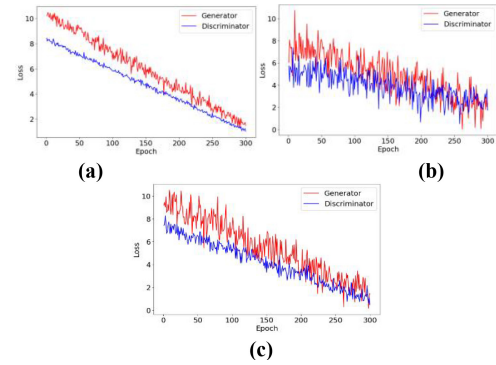


Fig. 11. Training loss curves for the generator and discriminator in the proposed GAN model over 300 epochs on the a) KDDcup99, b) ISCX-UNB, and c) CICDDOS datasets.

The proposed GAN produces more dispersed distributions that closely resemble real data, reflecting its effectiveness in avoiding mode collapse. Figure 11 shows the training loss curves of the generator and discriminator over 300 epochs on the KDDcup99, ISCX-UNB, and CICDDOS datasets. Both losses decline steadily, indicating stable convergence. The generator learns to produce realistic data, while the discriminator improves its accuracy.

TABLE VIII
PERFORMANCE METRICS COMPARISON OF VARIOUS HYPERPARAMETER OPTIMIZATION METHODS ACROSS THE KDDCUP99 DATASET

Method	Accuracy	Precision	Recall	F-measure	G-means	MSE
Random search	73.401 ± 0.059	70.423 ± 0.036	69.485 ± 0.095	70.073 ± 0.078	70.911 ± 0.099	7.712 ± 0.094
Grid search	75.067 ± 0.041	72.331 ± 0.043	71.218 ± 0.097	71.751 ± 0.027	72.545 ± 0.025	6.728 ± 0.003
Bayesian optimization	90.881 ± 0.006	88.566 ± 0.051	87.788 ± 0.083	88.073 ± 0.022	88.727 ± 0.081	1.898 ± 0.033
Hyperband	84.899 ± 0.084	84.204 ± 0.081	83.492 ± 0.037	83.765 ± 0.029	84.539 ± 0.025	2.853 ± 0.024
WSO	75.893 ± 0.019	73.117 ± 0.028	72.582 ± 0.019	72.855 ± 0.036	73.581 ± 0.089	5.745 ± 0.024
WOA	77.273 ± 0.002	74.268 ± 0.056	73.684 ± 0.072	74.006 ± 0.056	74.762 ± 0.095	5.750 ± 0.056
PO	78.260 ± 0.007	75.628 ± 0.063	74.995 ± 0.05	75.405 ± 0.019	76.092 ± 0.031	4.764 ± 0.084
CMA-ES	79.226 ± 0.055	77.155 ± 0.014	76.236 ± 0.021	76.609 ± 0.004	77.316 ± 0.035	3.781 ± 0.080
ABC	79.934 ± 0.063	78.024 ± 0.065	77.268 ± 0.055	77.584 ± 0.021	78.302 ± 0.063	2.794 ± 0.068
DE	81.043 ± 0.078	79.637 ± 0.025	78.709 ± 0.013	79.154 ± 0.048	79.855 ± 0.089	2.804 ± 0.093
BOUB	90.498 ± 0.095	90.983 ± 0.011	87.651 ± 0.031	89.286 ± 0.096	90.983 ± 0.047	1.142 ± 0.027

TABLE IX
PERFORMANCE METRICS COMPARISON OF VARIOUS HYPERPARAMETER OPTIMIZATION METHODS ACROSS THE ISCX-UNB DATASET

Method	Accuracy	Precision	Recall	F-measure	G-means	MSE
Random search	74.909 ± 0.077	72.256 ± 0.075	71.228 ± 0.054	71.747 ± 0.002	72.460 ± 0.070	6.642 ± 0.042
Grid search	76.200 ± 0.091	73.372 ± 0.065	72.554 ± 0.048	72.986 ± 0.051	73.643 ± 0.092	6.648 ± 0.073
Bayesian optimization	92.168 ± 0.085	90.364 ± 0.075	89.678 ± 0.038	89.996 ± 0.082	90.387 ± 0.082	1.126 ± 0.033
Hyperband	85.611 ± 0.012	84.38 ± 0.039	83.496 ± 0.048	83.915 ± 0.024	84.502 ± 0.075	1.760 ± 0.080
WSO	76.796 ± 0.009	74.759 ± 0.043	73.722 ± 0.09	74.243 ± 0.082	74.910 ± 0.029	4.663 ± 0.037
WOA	78.243 ± 0.065	75.965 ± 0.031	75.183 ± 0.095	75.538 ± 0.036	76.123 ± 0.020	4.679 ± 0.013
PO	79.610 ± 0.070	77.122 ± 0.036	76.175 ± 0.055	76.528 ± 0.098	77.146 ± 0.089	3.691 ± 0.077
CMA-ES	80.574 ± 0.063	77.972 ± 0.067	77.008 ± 0.066	77.467 ± 0.090	78.096 ± 0.017	3.699 ± 0.082
ABC	81.170 ± 0.071	79.421 ± 0.018	78.517 ± 0.02	78.977 ± 0.014	79.574 ± 0.079	2.709 ± 0.077
DE	82.251 ± 0.094	80.775 ± 0.049	79.867 ± 0.039	80.357 ± 0.094	80.903 ± 0.073	1.719 ± 0.002
BOUB	93.186 ± 0.025	91.902 ± 0.019	90.869 ± 0.065	91.196 ± 0.055	91.896 ± 0.050	0.942 ± 0.027

TABLE X
PERFORMANCE COMPARISON OF THE VARIOUS GANs FOR DATA AUGMENTATION USING THE CICDDOS DATASET

Model	FID	MMD	Wasserstein	KS
GAN	8.266 ± 0.096	11.566 ± 0.027	17.114 ± 0.014	0.592 ± 0.046
WGAN	8.488 ± 0.064	10.932 ± 0.057	15.224 ± 0.025	0.633 ± 0.033
SAGAN	6.645 ± 0.024	10.828 ± 0.036	13.758 ± 0.095	0.588 ± 0.026
PGGAN	6.973 ± 0.088	9.521 ± 0.023	12.873 ± 0.083	0.602 ± 0.01
BigGAN	4.505 ± 0.074	7.677 ± 0.043	10.325 ± 0.019	0.676 ± 0.039
StyleGAN	3.114 ± 0.097	4.224 ± 0.081	5.351 ± 0.05	0.598 ± 0.045
Proposed	1.327 ± 0.029	0.934 ± 0.056	1.407 ± 0.055	0.376 ± 0.046

4) *Analysis of BOUB*: This section evaluates BOHB against three traditional methods: random search, grid search, and Bayesian optimization, along with six evolutionary algorithms: white shark optimizer (WSO) [52], whale optimization algorithm (WOA) [53], puma optimizer

TABLE XI
PERFORMANCE METRICS COMPARISON OF VARIOUS HYPERPARAMETER OPTIMIZATION METHODS ACROSS THE CICDDOS DATASET

Method	Accuracy	Precision	Recall	F-measure	G-means	MSE
Random search	74.658 ± 0.096	71.845 ± 0.043	70.746 ± 0.074	71.029 ± 0.082	73.234 ± 0.052	6.222 ± 0.025
Grid search	76.664 ± 0.064	73.284 ± 0.081	72.177 ± 0.057	72.135 ± 0.099	72.882 ± 0.074	7.507 ± 0.083
Bayesian optimization	91.480 ± 0.024	89.763 ± 0.056	89.771 ± 0.027	90.540 ± 0.028	90.909 ± 0.061	1.393 ± 0.088
Hyperband	84.727 ± 0.088	84.565 ± 0.014	84.435 ± 0.082	82.926 ± 0.083	85.044 ± 0.054	2.367 ± 0.027
WSO	76.998 ± 0.074	74.974 ± 0.025	74.601 ± 0.091	74.657 ± 0.076	74.955 ± 0.048	5.448 ± 0.059
WOA	77.284 ± 0.097	75.095 ± 0.095	75.379 ± 0.093	76.081 ± 0.017	75.174 ± 0.020	5.294 ± 0.091
PO	80.275 ± 0.029	78.053 ± 0.083	75.352 ± 0.028	76.245 ± 0.020	76.209 ± 0.067	3.327 ± 0.020
CMA-ES	79.938 ± 0.027	77.581 ± 0.019	76.098 ± 0.039	78.193 ± 0.066	77.725 ± 0.056	3.155 ± 0.048
ABC	80.778 ± 0.057	79.789 ± 0.050	78.294 ± 0.034	78.639 ± 0.016	80.389 ± 0.032	3.345 ± 0.087
DE	82.115 ± 0.036	80.019 ± 0.055	80.524 ± 0.042	79.979 ± 0.039	80.724 ± 0.078	0.733 ± 0.056
BOUB	92.403 ± 0.080	89.652 ± 0.056	89.970 ± 0.032	88.712 ± 0.070	90.285 ± 0.019	0.557 ± 0.017

TABLE XII
COMPARATIVE ANALYSIS OF COMPUTATIONAL EFFICIENCY ACROSS VARIOUS HYPERPARAMETER OPTIMIZATION ALGORITHMS ON THE KDDCUP99, ISCX-UNB, AND CICDDOS DATASETS

ALGORITHM	KDDcup99		ISCX-UNB		CICDDOS	
	Runtime	GPU	Runtime	GPU	Runtime	GPU
Random search	2267	20.5	2060	19.4	2140	19.8
Grid search	2346	22.8	2156	20.1	2205	22.9
Bayesian optimization	2965	20.2	2836	19.3	2875	18.8
Hyperband	2026	18.7	1962	17.6	2036	17.9
WSO	3125	21.6	2866	20.1	3075	20.5
WOA	3256	22.4	3365	20.7	3204	21.3
PO	3352	25.6	3167	24.9	3296	24.9
CMA-ES	3105	23.9	3056	21.4	3075	22.0
ABC	2963	22.5	2752	20.0	2804	21.6
DE	2872	21.4	2683	20.7	2703	20.9
BOUB	2726	19.6	2587	18.4	2659	18.7

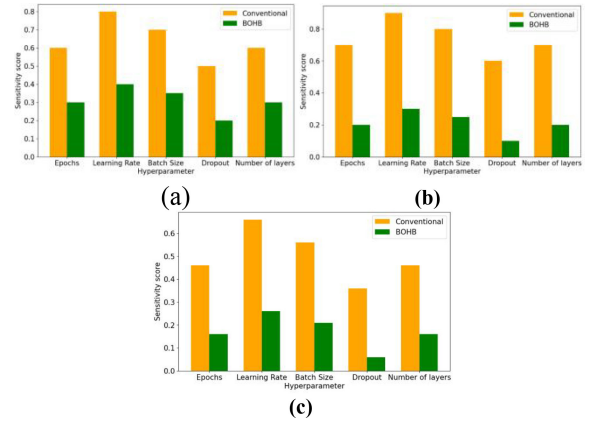


Fig. 12. Hyperparameter sensitivity analysis of BOHB versus conventional tuning on the a) KDDcup99, b) ISCX-UNB, and c) CICDDOS datasets.

(PO) [54], covariance matrix adaptation evolution strategy (CMA-ES) [55], artificial bee colony (ABC) [56], and differential evolution (DE) [56]. All model variables were standardized for fair comparison. As shown in Tables VII, IX, and X, BOHB achieves faster convergence and strong performance across the KDDcup99, ISCX-UNB, and CICDDOS datasets. Its effective balance between exploration and exploitation demonstrates robustness and efficiency for real-world hyperparameter tuning tasks.

TABLE XIII
PERFORMANCE COMPARISON OF THE PROPOSED MODEL AGAINST ML AND DL MODELS USING THE DARPA DATASET

Approach	Model	Accuracy	Precision	Recall	F-measure	G-means	MSE
ML	DPTCM-KNN [41]	72.132 ± 0.001	69.525 ± 0.049	69.045 ± 0.053	69.208 ± 0.054	70.012 ± 0.023	7.262 ± 0.086
	ERF-KMC [42]	73.674 ± 0.037	71.157 ± 0.047	71.939 ± 0.012	70.369 ± 0.016	71.180 ± 0.098	7.211 ± 0.005
	FSSVM [1]	74.187 ± 0.047	71.663 ± 0.022	71.238 ± 0.019	71.327 ± 0.040	72.102 ± 0.087	6.675 ± 0.015
	ML-CFS [6]	74.708 ± 0.045	71.812 ± 0.049	72.672 ± 0.017	72.141 ± 0.092	72.955 ± 0.001	6.262 ± 0.081
	ML-EA [43]	75.455 ± 0.015	72.693 ± 0.032	73.445 ± 0.061	73.081 ± 0.038	73.943 ± 0.083	6.632 ± 0.003
DL	IDAD [23]	76.005 ± 0.049	74.152 ± 0.044	75.127 ± 0.072	74.426 ± 0.087	75.182 ± 0.087	6.056 ± 0.045
	SDN-Edge-IoT [24]	77.442 ± 0.017	75.109 ± 0.027	76.818 ± 0.038	75.439 ± 0.053	76.203 ± 0.045	5.142 ± 0.012
	CNN-RPL [29]	79.956 ± 0.034	79.425 ± 0.042	80.612 ± 0.06	79.506 ± 0.054	80.238 ± 0.037	6.036 ± 0.099
	CNN-BiLSTM [44]	80.707 ± 0.014	80.498 ± 0.084	81.123 ± 0.086	80.711 ± 0.082	81.397 ± 0.023	5.142 ± 0.017
	ATL [30]	81.844 ± 0.003	81.245 ± 0.098	82.692 ± 0.074	81.522 ± 0.064	82.202 ± 0.068	5.685 ± 0.043
	LSTM-CNN [45]	82.626 ± 0.040	82.814 ± 0.088	84.686 ± 0.097	82.941 ± 0.081	83.642 ± 0.062	4.122 ± 0.025
	MDDCC [46]	83.790 ± 0.059	84.178 ± 0.019	85.965 ± 0.01	84.502 ± 0.041	85.194 ± 0.029	4.162 ± 0.022
	DTL-5G [31]	84.328 ± 0.049	84.853 ± 0.081	86.862 ± 0.075	85.470 ± 0.007	86.124 ± 0.043	3.126 ± 0.074
	FCNN [3]	85.521 ± 0.067	86.774 ± 0.03	86.982 ± 0.09	86.972 ± 0.011	87.603 ± 0.092	3.185 ± 0.086
	mGRU [47]	87.007 ± 0.051	87.343 ± 0.029	87.854 ± 0.037	87.608 ± 0.071	88.233 ± 0.029	2.362 ± 0.067
RL	DDoSViT [32]	87.626 ± 0.041	84.745 ± 0.068	85.132 ± 0.01	84.967 ± 0.092	85.801 ± 0.088	3.180 ± 0.086
	MD-RL [21]	77.038 ± 0.044	74.396 ± 0.035	75.468 ± 0.019	74.542 ± 0.078	75.313 ± 0.039	4.671 ± 0.080
	Q-learning [34]	85.852 ± 0.050	84.077 ± 0.036	84.106 ± 0.045	84.036 ± 0.076	84.852 ± 0.046	3.770 ± 0.074
	MEDRL [35]	87.626 ± 0.041	84.866 ± 0.025	84.985 ± 0.054	84.967 ± 0.092	85.801 ± 0.088	3.789 ± 0.086
	RL-ARL [36]	89.438 ± 0.067	86.054 ± 0.058	86.207 ± 0.048	86.137 ± 0.016	86.998 ± 0.027	2.792 ± 0.037
Ablation	Proposed w/o DA	87.117 ± 0.015	85.325 ± 0.071	85.561 ± 0.057	85.437 ± 0.004	86.990 ± 0.072	1.864 ± 0.008
	Proposed w/o FS	83.704 ± 0.012	80.482 ± 0.098	80.786 ± 0.062	80.591 ± 0.062	81.193 ± 0.021	2.172 ± 0.077
	Proposed w/o off-policy PPO	85.331 ± 0.097	82.231 ± 0.019	82.914 ± 0.051	82.744 ± 0.053	81.348 ± 0.079	1.979 ± 0.054
	Proposed w/o HO	89.493 ± 0.022	87.207 ± 0.027	87.913 ± 0.048	87.339 ± 0.082	86.901 ± 0.019	1.790 ± 0.033
	Proposed using only PPO + SHAP	90.568 ± 0.042	88.042 ± 0.09	88.839 ± 0.059	88.426 ± 0.012	87.565 ± 0.025	1.345 ± 0.045
	Proposed model	92.849 ± 0.040	92.039 ± 0.039	92.491 ± 0.099	92.192 ± 0.072	92.790 ± 0.007	1.102 ± 0.050

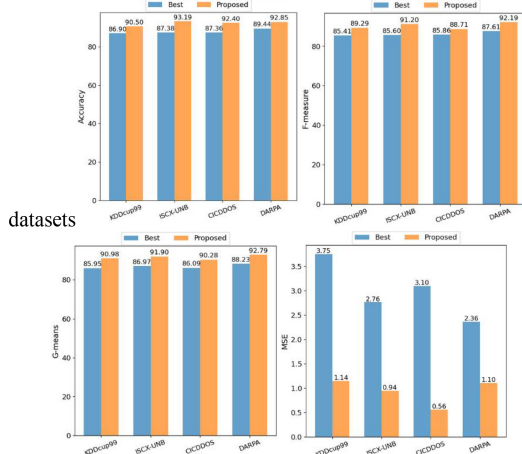


Fig. 13. Comparative performance of the proposed model vs. best-existing models on the KDDcup99, ISCX-UNB, DARPA, and CICDDoS datasets.

Figure 12 shows that BOHB achieves lower hyperparameter sensitivity than traditional methods on the ISCX UNB and KDDCup99 datasets, indicating greater model stability. Key improvements are seen in learning rate, batch size, dropout, and number of layers. BOHB effectively reduces overfitting and enhances generalization, making it suitable for complex datasets. Figure 13 compares the proposed model with top existing methods on the KDDcup99, ISCX UNB, and DARPA datasets using Accuracy, F-measure, G-means, and MSE. The proposed model consistently outperforms others, showing notable accuracy gains across all datasets.

V. CONCLUSION

This research proposes a novel solution for DDoS attack detection in healthcare by integrating off-policy PPO, an

improved GAN, and the BOHB algorithm. The approach effectively addresses imbalanced data and hyperparameter tuning, which are key challenges in real-world detection. Off-policy PPO improves classification on imbalanced data using experience updates and SHAP-based feature selection. The enhanced GAN with selective gradient exclusion generates diverse and relevant samples, increasing model robustness.

BOHB accelerates hyperparameter tuning, which is essential for time-sensitive healthcare environments. Experiments on the KDDcup99, ISCX-UNB, CICDDoS, and DARPA datasets confirm the model's high accuracy and F-measure, highlighting its potential to strengthen DDoS detection and safeguard healthcare systems. Moreover, while the proposed system is tailored to consumer IoT in healthcare, its modular architecture and reliance on packet-level features make it adaptable to other IoT-based domains such as smart cities and industrial IoT. With minor adjustments in feature engineering and retraining, the framework can generalize well to diverse deployment scenarios.

REFERENCES

- [1] A. V. Chandak and N. K. Ray, "DDoS attack detection in smart home applications," *Softw., Pract. Exp.*, vol. 54, no. 10, pp. 2086–2101, 2024.
- [2] J. Deng et al., "FIDSUS: Federated intrusion detection for securing UAV swarms in smart aerial computing," *IEEE Internet Things J.*, vol. 12, no. 9, pp. 11312–11328, May 2025, doi: [10.1109/jiot.2025.3549508](https://doi.org/10.1109/jiot.2025.3549508).
- [3] A. Bhatt, A. Gohil, S. Vaghasiya, R. Gupta, S. Tanwar, and J. Bhatia, "DL-based DDoS attack detection in SDN-assisted healthcare 4.0 telesurgery networks," in *Proc. 5th Int. Conf. Emerg. Technol. (INCET)*, 2024, pp. 1–7.
- [4] M. Wazid, A. K. Das, and S. Shetty, "BSFR-SH: Blockchain-enabled security framework against ransomware attacks for smart healthcare," *IEEE Trans. Consum. Electron.*, vol. 69, no. 1, pp. 18–28, Feb. 2023.

- [5] P. P. Roy, V. Teju, S. R. Kandula, K. V. Sowmya, A. I. Stan, and O. P. Stan, "Secure healthcare model using multi-step deep Q learning network in Internet of Things," *Electronics*, vol. 13, no. 3, p. 669, 2024.
- [6] E. Altulaihan, M. A. Almaiah, and A. Aljughaiman, "Anomaly detection IDS for detecting DoS attacks in IoT networks based on machine learning algorithms," *Sensors*, vol. 24, no. 2, p. 713, 2024.
- [7] M. F. Saiyed, I. Al-Anbagi, and M. S. Hossain, "Interactive and explainable optimized learning for DDoS detection in consumer IoT networks," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 6839–6853, May 2025, doi: [10.1109/TCE.2024.3482092](https://doi.org/10.1109/TCE.2024.3482092).
- [8] J. Manokaran and G. Vairavel, "GIWRF-SMOTE: Gini impurity-based weighted random forest with SMOTE for effective malware attack and anomaly detection in IoT-Edge," *Smart Sci.*, vol. 11, no. 2, pp. 276–292, 2023.
- [9] T. Sun, X. Zhang, X. Wang, Y. Zhuang, and Z. He, "Traffic intrusion detection of medical consumption electronics in the field of medical management based on integrated learning" *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 1334–1341, Feb. 2024.
- [10] R. Jiao, C. Li, G. Xun, T. Zhang, B. B. Gupta, and G. Yan, "A context-aware multi-event identification method for nonintrusive load monitoring," *IEEE Trans. Consum. Electron.*, vol. 69, no. 2, pp. 194–204, May 2023, doi: [10.1109/TCE.2023.3236452](https://doi.org/10.1109/TCE.2023.3236452).
- [11] K. Li, Z. Dai, X. Wang, Y. Song, and G. Jeon, "GAN-based controllable image data augmentation in low-visibility conditions for improved roadside traffic perception," *IEEE Trans. Consum. Electron.*, vol. 70, no. 3, pp. 6174–6188, Aug. 2024.
- [12] C. Zhan, D. Yin, Y. Shen, and T. Hao, "GMINN: A generative moving interactive neural network for enhanced short-term load forecasting in modern electricity markets," *IEEE Trans. Consum. Electron.*, vol. 70, no. 3, pp. 5461–5470, Aug. 2024.
- [13] L. P. S. Rao and S. Jaganathan, "Adaptive Bayesian contextual hyperband: A novel hyperparameter optimization approach," *Int. J. Artif. Intell.*, vol. 13, no. 1, pp. 775–785, 2023.
- [14] M. Cao, T. Zhang, Y. Liu, Y. Wang, and Z. Shi, "A Bayesian optimization hyperband-optimized incremental deep belief network for online battery behaviour modelling for a satellite simulator," *J. Energy Storage*, vol. 58, Feb. 2023, Art. no. 106348.
- [15] Y. Feng, J. Li, D. Sisodia, and P. Reiher, "On explainable and adaptable detection of distributed denial-of-service traffic," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 4, pp. 2211–2226, Jul./Aug. 2024.
- [16] D. Tang, Y. Yan, C. Gao, W. Liang, and W. Jin, "LrFT: Mitigate the low-rate data plane DDoS attack with learning-to-rank enabled flow tables," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 3143–3157, 2023.
- [17] Y. Xu, S. Zhang, C. Lyu, J. Liu, Y. Shen, and S. Norio, "Mitigating distributed DoS attacks on bandwidth allocation for federated learning in mobile edge networks," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 3, pp. 1941–1960, May/Jun. 2025.
- [18] Z. Zhao, Z. Liu, H. Chen, F. Zhang, Z. Song, and Z. Li, "Effective DDoS mitigation via ML-driven in-network traffic shaping," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 4, pp. 4271–4289, Jul./Aug. 2024.
- [19] S. M. M. Mirnajafizadeh, A. R. Sethuram, D. Mohaisen, D. Nyang, and R. Jang, "Enhancing network attack detection with distributed and in-network data collection system," in *Proc. 33rd USENIX Secur. Symp. (USENIX Secur.)*, 2024, pp. 5161–5178.
- [20] H. Setia et al., "Securing the road ahead: Machine learning-driven DDoS attack detection in VANET cloud environments," *Cyber Secur. Appl.*, vol. 2, Jan. 2024, Art. no. 100037.
- [21] S. Mahato, M. S. Obaidat, S. Dutta, D. Giri, and M. S. Hossain, "Enhancing malicious activity detection in IoT-enabled network and IoMT systems through meta-heuristic optimization and machine learning," *IEEE Internet Things J.*, vol. 11, no. 24, pp. 39958–39970, Dec. 2024.
- [22] Q. Fan et al., "IDAD: An improved tensor train based distributed DDoS attack detection framework and its application in complex networks," *Future Gener. Comput. Syst.*, vol. 162, Jan. 2025, Art. no. 107471.
- [23] H. M. Belachew, M. Y. Beyene, A. B. Desta, B. T. Alemu, S. S. Musa, and A. J. Muhammed, "Design a robust DDoS attack detection and mitigation scheme in SDN-edge-IoT by leveraging machine learning," *IEEE Access*, vol. 13, pp. 10194–10214, 2025.
- [24] K. J. Pradeep and P. K. Shukla, "Designing a novel network anomaly detection framework using multi-serial stacked network with optimal feature selection procedures over DDOS attacks," *Int. J. Intell. Netw.*, vol. 6, pp. 1–13, Jan. 2025.
- [25] M. Dilshad, M. H. Syed, and S. Rehman, "Efficient distributed denial of service attack detection in Internet of Vehicles using Gini index feature selection and federated learning," *Future Internet*, vol. 17, no. 1, p. 9, 2025.
- [26] Y. Hosain and M. Çakmak, "XAI-XGBoost: An innovative explainable intrusion detection approach for securing internet of medical things systems," *Sci. Rep.*, vol. 15, no. 1, 2025, Art. no. 22278.
- [27] C. Benzaïd, T. Taleb, A. Sami, and O. Hireche, "FortisEDoS: A deep transfer learning-empowered economical denial of sustainability detection framework for cloud-native network slicing," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 4, pp. 2818–2835, Jul./Aug. 2024.
- [28] G. Duan, H. Lv, H. Wang, G. Feng, and X. Li, "Practical cyber attack detection with continuous temporal graph in dynamic network system," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 4851–4864, 2024.
- [29] C.-S. Shieh, F.-A. Ho, M.-F. Horng, T.-T. Nguyen, and P. Chakrabarti, "Open-set recognition in unknown DDoS attacks detection with reciprocal points learning," *IEEE Access*, vol. 12, pp. 56461–56476, 2024.
- [30] M. B. Anley, A. Genovese, D. Agostinello, and V. Piuri, "Robust DDoS attack detection with adaptive transfer learning," *Comput. Secur.*, vol. 144, Sep. 2024, Art. no. 103962.
- [31] B. Farzaneh, N. Shahriar, A. H. Al Muktadir, M. S. Towhid, and M. S. Khosravani, "DTL-5G: Deep transfer learning-based DDoS attack detection in 5G and beyond networks," *Comput. Commun.*, vol. 228, Dec. 2024, Art. no. 107927.
- [32] M. Ali, Y. Saleem, S. Hina, and G. A. Shah, "DDoSViT: IoT DDoS attack detection for fortifying firmware Over-The-Air (OTA) updates using vision transformer," *Internet Things*, vol. 30, Mar. 2025, Art. no. 101527.
- [33] Z. Ullah et al., "Hybrid CNN-LSTM model for DDoS attack detection in Internet of Things-based healthcare industry 5.0," *IEEE Internet Things J.*, early access, May 13, 2025, doi: [10.1109/JIOT.2025.3569610](https://doi.org/10.1109/JIOT.2025.3569610).
- [34] J. Doménech, O. León, M. S. Siddiqui, and J. Pegueroles, "Evaluating and enhancing intrusion detection systems in IoMT: The importance of domain-specific datasets," *Internet Things*, vol. 32, Jul. 2025, Art. no. 101631.
- [35] A. Kumar and D. Singh, "Detection and prevention of DDoS attacks on edge computing of IoT devices through reinforcement learning," *Int. J. Inf. Technol.*, vol. 16, no. 3, pp. 1365–1376, 2024.
- [36] D. Said, M. Bagaa, A. Oukaira, and A. Lakhssassi, "Quantum entropy and reinforcement learning for distributed denial of service attack detection in smart grid," *IEEE Access*, vol. 12, pp. 129858–129869, 2024.
- [37] N. Jayakrishna and N. N. Prasanth, "Detection and mitigation of distributed denial of service attacks in vehicular ad hoc network using a spatiotemporal deep learning and reinforcement learning approach," *Results Eng.*, vol. 26, Jun. 2025, Art. no. 104839.
- [38] K. Gai, G. Zhang, P. Jiang, L. Zhu, and K.-K. R. Choo, "A moving target DDoS defense approach in consortium blockchain," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 4, pp. 4362–4378, Jul./Aug. 2025.
- [39] K. K. Paidipati, C. Kurangi, J. Uthayakumar, S. Padmanayaki, D. Pradeepa, and S. Nithinsha, "Ensemble of deep reinforcement learning with optimization model for DDoS attack detection and classification in cloud based software defined networks," *Multimedia Tools Appl.*, vol. 83, no. 11, pp. 32367–32385, 2024.
- [40] J. A. Shaikh et al., "A deep reinforcement learning-based robust intrusion detection system for securing IoMT healthcare networks," *Front. Med.*, vol. 12, Apr. 2025, Art. no. 1524286.
- [41] Y. Geng, Y. Du, and J. Li, "The support vector machine model and application based on particle swarm optimization algorithm," in *Proc. IEEE 2nd Int. Conf. Image Process. Comput. Appl. (ICIPCA)*, 2024, pp. 898–902.
- [42] A. Farah, M. Nielsen, and E. Vasilomanolakis, "Improving synthetic network attack traffic generation," in *Proc. 9th Int. Workshop Traffic Meas. Cybersecur.*, 2024, pp. 326–334.
- [43] D. Kapil, V. Mittal, and D. P. Gangodkar, "Evaluating machine learning approaches for DDoS attack detection using CIC-DDoS2019," in *Proc. 2nd Int. Conf. Adv. Comput. Commun. Technol. (ICACCTech)*, 2024, pp. 762–767.

- [44] W. Mason, "The future of infrared efforts at DARPA," in *Proc. Infrared Technol. Appl.*, 2024, Art. no. 130460U.
- [45] G. Kaur and P. Gupta, "Detection of distributed denial of service attacks for IoT-based healthcare systems," *Comput. Assist. Methods Eng. Sci.*, vol. 30, no. 2, pp. 167–186, 2023.
- [46] A. A. J. Al-Abadi, M. B. Mohamed, and A. Fakhfakh, "Enhanced random forest classifier with K-means clustering (ERF-KMC) for detecting and preventing distributed-denial-of-service and man-in-the-middle attacks in Internet-of-Medical-Things networks," *Computers*, vol. 12, no. 12, p. 262, 2023.
- [47] F. Talpur, I. A. Korejo, A. A. Chandio, A. Ghulam, and M. S. H. Talpur, "ML-based detection of DDoS attacks using evolutionary algorithms optimization," *Sensors*, vol. 24, no. 5, p. 1672, 2024.
- [48] I. Jebril et al., "Deep learning based DDoS attack detection in Internet of Things: An optimized CNN-BiLSTM architecture with transfer learning and regularization techniques," *Infocommun. J.*, vol. 16, no. 1, pp. 2–11, 2024.
- [49] B. Habib and F. Khursheed, "Time-based DDoS attack detection through hybrid LSTM-CNN model architectures: An investigation of many-to-one and many-to-many approaches," *Concurr. Comput., Pract. Exp.*, vol. 36, no. 9, 2024, Art. no. e7996.
- [50] K. Wang, Y. Fu, X. Duan, and T. Liu, "Detection and mitigation of DDoS attacks based on multi-dimensional characteristics in SDN," *Sci. Rep.*, vol. 14, no. 1, 2024, Art. no. 16421.
- [51] A. D. Aguru and S. B. Erukala, "A lightweight multi-vector DDoS detection framework for IoT-enabled mobile health informatics systems using deep learning," *Inf. Sci.*, vol. 662, Mar. 2024, Art. no. 120209.
- [52] M. Braik, A. Hammouri, J. Atwan, M. A. Al-Betar, and M. A. Awadallah, "White shark optimizer: A novel bio-inspired metaheuristic algorithm for global optimization problems," *Knowl. Based Syst.*, vol. 243, May 2022, Art. no. 108457.
- [53] M. Jafari, M. H. B. Chaleshtari, H. Khoramshad, and H. Altenbach, "Minimization of thermal stress in perforated composite plate using metaheuristic algorithms WOA, SCA and GA," *Composite Struct.*, vol. 304, Jan. 2023, Art. no. 116403.
- [54] B. Abdollahzadeh et al., "Puma optimizer (PO): A novel metaheuristic optimization algorithm and its application in machine learning," *Clust. Comput.*, vol. 27, pp. 5235–5283, Jan. 2024.
- [55] N. Sinha and K.-W. Chen, "Neural architecture search using covariance matrix adaptation evolution strategy," *Evol. Comput.*, vol. 32, no. 2, pp. 177–204, 2024. [Online]. Available: https://doi.org/10.1162/evco_a_00331
- [56] S. V. Moravvej, S. J. Mousavirad, D. Oliva, G. Schaefer, and Z. Sobhaninia, "An improved de algorithm to optimise the learning process of a BERT-based plagiarism detection model," in *Proc. IEEE Congr. Evol. Comput. (CEC)*, 2022, pp. 1–7.